Università degli studi di Milano-Bicocca

Data Science Lab for Smart Cities

Final Essay

Equity in Cycling Mobility: A Data-Driven Analysis of BikeMi and Cycling Infrastructure in Milan

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May 28, 2026



Abstract

This paper investigates whether access to cycling mobility in Milan — specifically to the BikeMi bike-sharing system and the cycling lane network — is equitably distributed across the city's 88 *Nuclei di Identità Locale* (NIL), or whether it follows a socioeconomic gradient that systematically disadvantages lower-income neighbourhoods. Four open datasets were integrated: BikeMi stations (325), cycling lane shapefile (4,154 segments, 332.8 km), NIL polygons (88), and official IRPEF income data for tax year 2022 (Comune di Milano, dataset DS2734). Spatial analysis, isochrone modelling, and correlation analysis reveal a strong positive correlation between neighbourhood income and BikeMi station density (Pearson $r = +0.716$, $p < 0.001$), while total cycling lane length shows no significant income gradient ($r = +0.052$, n.s.). A Decision Tree classifier (5-fold CV accuracy = 79.0%) identifies 16 priority NILs for new BikeMi station investment, predominantly peripheral low-income areas. Four policy recommendations are formulated: targeted BikeMi expansion, cycling infrastructure quality upgrades, equity-adjusted pricing, and an annual monitoring framework.

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1 Problem Description, Literature Review and Indicators

1.1 Introduction

Sustainable urban mobility has become a central priority for European cities over the past two decades. Cycling, in particular, has been widely promoted as a low-carbon, space-efficient, and health-promoting mode of transport. Yet the expansion of cycling infrastructure does not automatically translate into equitable access for all urban residents. In many cities, cycling infrastructure and bike-sharing services tend to concentrate in wealthier, central neighbourhoods, reinforcing rather than reducing existing spatial inequalities in mobility.

This paper investigates whether access to cycling mobility in Milan — specifically to the BikeMi bike-sharing system and the cycling lane network — is equitably distributed across the city's 88 NILs, or whether it follows a socioeconomic gradient that systematically disadvantages lower-income neighbourhoods. The question has direct implications for urban justice, sustainable transport planning, and the effectiveness of Milan's Sustainable Urban Mobility Plan (PUMS) [1].

1.2 Societal Significance

Cycling mobility is not a neutral public good. Research consistently shows that in cities with growing cycling networks, access to safe cycling infrastructure and bike-sharing stations tends to be unevenly distributed along income lines. This "cycling equity gap" matters for several reasons.

First, lower-income residents often have the greatest need for affordable transportation alternatives, having less access to private vehicles and being more sensitive to public transport costs. Second, cycling infrastructure in low-income areas tends to be of lower quality — shared with pedestrians or vehicles rather than physically separated — making it less safe and less attractive. Third, bike-sharing systems, despite being designed as democratic mobility tools, frequently suffer from what the literature calls "first-mile and last-mile deserts" in peripheral, lower-income areas.

Milan is an ideal case study. Its PUMS explicitly lists equity, security, and social inclusion among its strategic objectives. The city operates BikeMi, one of Italy's largest bike-sharing systems, with 325 stations and over 4,700 docks. It has invested significantly in cycling infrastructure, reaching 332.8 km of cycling lanes. Yet, as this analysis will demonstrate, the distribution of these resources is far from uniform across the city's neighbourhoods.

1.3 Literature Review

1.3.1 Transport Equity: Theoretical Frameworks

Equity in transport refers to the fair distribution of transport resources and opportunities among different societal groups. The scientific literature distinguishes two complementary dimensions [2]: **horizontal equity** (spatial fairness — are resources evenly distributed across geographical areas?) and **vertical equity** (social fairness — do vulnerable populations receive adequate services given their higher dependence on public mobility?).

Schwanen [3] identifies three key dimensions of transport injustice: exposure to traffic risks and pollutants, distribution of public space, and unequal valuation of transport time. Lower-income populations tend to fare worse on all three. The development of smart city mobility systems can further exacerbate these inequalities, creating what the literature calls "sensor deserts" and "mobility divides" that disproportionately impact vulnerable populations [4].

1.3.2 Bike-Sharing and Equity

Research on bike-sharing equity has grown substantially over the past decade. Several studies have documented a consistent pattern: bike-sharing stations tend to cluster in high-income, central urban areas, while peripheral lower-income neighbourhoods are systematically underserved. Ogilvie and Goodman [5] showed that cycling uptake following infrastructure investment benefited primarily higher-income groups. Ursaki and Aultman-Hall [6] found that in US cities, bike-share programs disproportionately served high-income populations. More recently, Qian and Jaller [7] demonstrated that in Chicago, bike-share access correlated positively with median household income at the neighbourhood level.

In the European context, the pattern is similar but modulated by stronger public transport networks. Studies in Barcelona, Paris, and Amsterdam have shown that while bike-sharing systems are more spatially integrated into public transport in European cities, income-based disparities in cycling infrastructure quality persist.

1.3.3 Cycling Infrastructure Quality

Not all cycling infrastructure is equal. The distinction between dedicated lanes (physically separated from traffic), signage-only lanes, and shared spaces is crucial for both safety and usage levels. Research by Pucher and Buehler [8] demonstrated that physical separation from motor traffic is the most significant predictor of cycling adoption, particularly among women, older adults, and lower-income cyclists. This finding is highly relevant to Milan, where 21.1% of cycling infrastructure is shared with motor vehicles — a category that effectively provides little safety benefit to cyclists.

1.3.4 Isochrone-Based Accessibility

Isochrone analysis — measuring the area reachable within a fixed travel time — has emerged as a standard methodology for assessing transport accessibility. Páez et al. [9] propose that pedestrian isochrones provide a more realistic measure of effective bike-sharing access than simple Euclidean buffer distances, since actual walking time depends on street network topology rather than straight-line distance. This methodology is adopted in the present analysis to measure the proportion of each NIL accessible within a 5-minute walk from a BikeMi station.

1.4 Indicators

Based on the literature and the specific context of Milan, this study employs five core indicators, computed at the NIL level:

1. **BikeMi Station Density** (stations/km^2): measures the intensity of bike-sharing provision per unit area. Its rationale follows from the literature on proximity as a determinant of bike-sharing usage [10].
2. **Cycling Lane Density** (km/km^2): measures the intensity of cycling infrastructure per unit area. Combined with station density, it captures whether stations are embedded in a supportive cycling network.
3. **Pedestrian Isochrone Coverage** (%): the proportion of a NIL's area reachable within 5 minutes on foot from the nearest BikeMi station, computed using the actual street network via OSMnx.
4. **Cycling Lane Quality Index** (% high-quality lanes): the proportion of cycling infrastructure classified as *ciclabile sede propria* (dedicated lane, physically separated from traffic).
5. **Average IRPEF Income** (€/taxpayer): the average taxable income per taxpayer in each NIL, computed from official Comune di Milano IRPEF data (dataset DS2734, tax year 2022), aggregated from CAP (postcode) to NIL level via weighted spatial join.

1.5 Ethical and Social Implications

The selection of income as an indicator carries intrinsic ethical implications. IRPEF data captures declared taxable income, which systematically underestimates the wealth of high-income households with substantial capital income or tax optimisation strategies. Conversely, it may overestimate the economic vulnerability of lower-income NILs that include significant informal economic activity. These limitations are acknowledged in the interpretation of results.

More broadly, the documentation of a positive correlation between income and cycling accessibility raises the question of whether public investments in cycling infrastructure function as a regressive redistribution mechanism — directing public resources toward already privileged neighbourhoods. If BikeMi expansion and cycling lane investment consistently favour high-income areas, the public subsidy implicit in these investments accrues disproportionately to wealthier residents.

The Decision Tree model developed in Part 2 classifies neighbourhoods as "priority" for new BikeMi stations based on income and cycling infrastructure quality. Using income as a model feature raises the ethical question of whether public infrastructure decisions should explicitly target low-income areas. We argue that doing so is a form of positive discrimination fully consistent with the vertical equity principle [11] and explicitly mandated by Milan's PUMS.

2 Data Analytics and Policy Suggestions

2.1 Datasets and Data Wrangling

Four datasets were integrated for this analysis, all from open data sources:

BikeMi Stations: 325 stations with geographic coordinates, number of docks, opening year, and administrative zone. Source: Comune di Milano Open Data portal. Numeric fields (year, docks, zone) were parsed with error coercion and rows with missing coordinates were dropped (2 rows).

Cycling Lane Shapefile: 4,154 lane segments with type (*tipologia*), length, completion year (*fine_lavor*), and geometric representation. Source: Comune di Milano Open Data portal (bike_ciclabili_shp.zip). All layers were reprojected to EPSG:32632 (UTM 32N) for metric calculations.

NIL Polygons: 88 neighbourhood polygons (*Nuclei di Identità Locale*) in GeoJSON format. Source: OpenData Milano GitHub mirror, originally derived from Comune di Milano administrative data.

IRPEF Income Data 2022: Dataset DS2734 from dati.comune.milano.it, containing total taxable income and number of taxpayers by CAP (postal code) for tax year 2022 [12]. Average income per CAP was computed as:

$$\text{avg_income} = \frac{\text{total_taxable_income}}{\text{n_taxpayers}}$$

CAP centroids were mapped to NIL polygons via spatial join, and NIL-level income was computed as a weighted average of CAP incomes, weighted by number of taxpayers. All spatial operations were performed using GeoPandas. The final analysis dataset comprises 32 NILs with complete data across all five indicators.

2.2 Temporal Analysis

Before examining the spatial distribution of cycling resources, we analyse their temporal evolution to assess whether BikeMi expansion and cycling infrastructure development have proceeded in a coordinated manner.

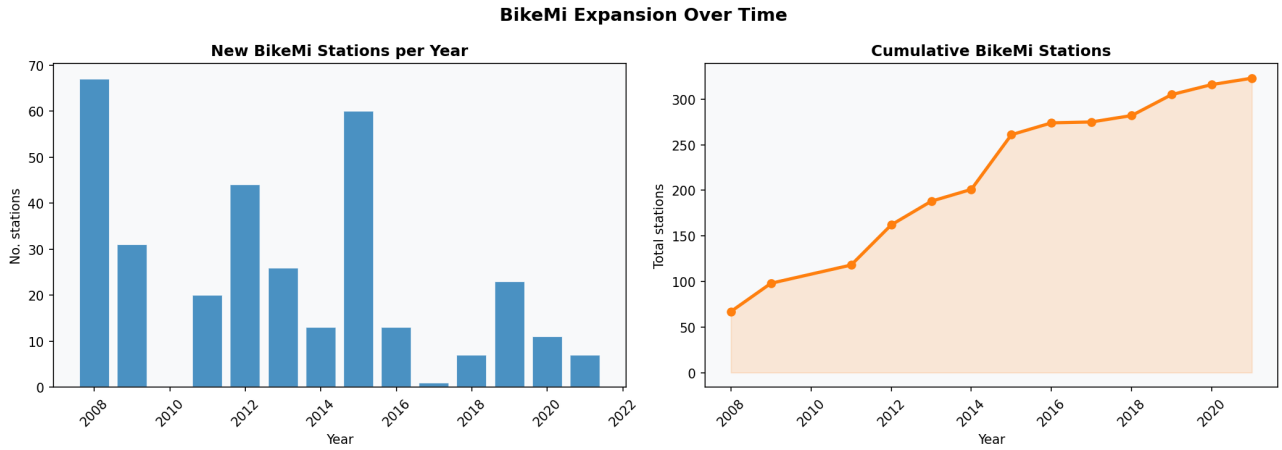


Figure 1: BikeMi expansion over time. Left: new stations opened per year (2008–2021). Right: cumulative growth curve. The system grew rapidly in its launch years (2008: 67 stations, 2015: 60), then entered a consolidation phase after 2016.

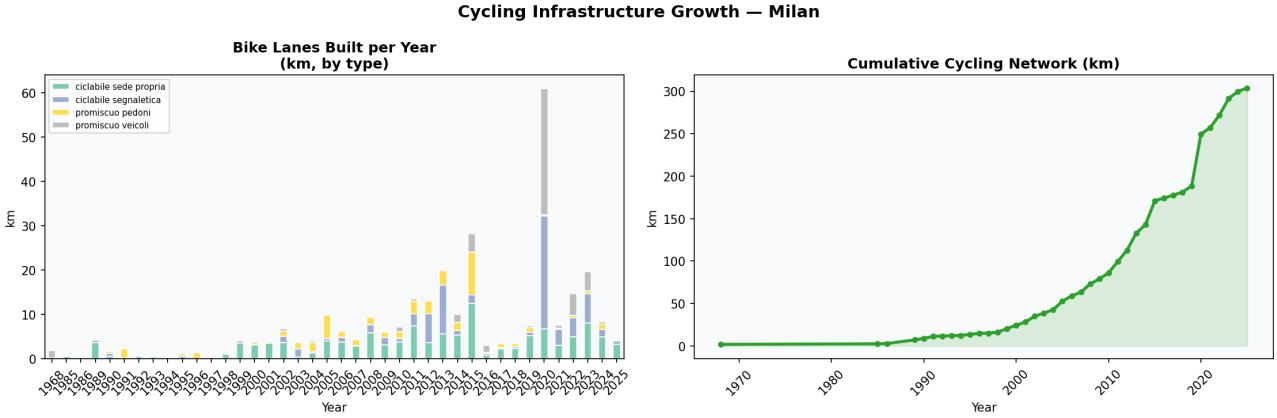


Figure 2: Cycling infrastructure growth over time. Left: km of lanes built per year by type. Right: cumulative network growth (km). A major acceleration occurred in 2020 under Milan's *Strade Aperte* emergency cycling programme during the COVID-19 pandemic.

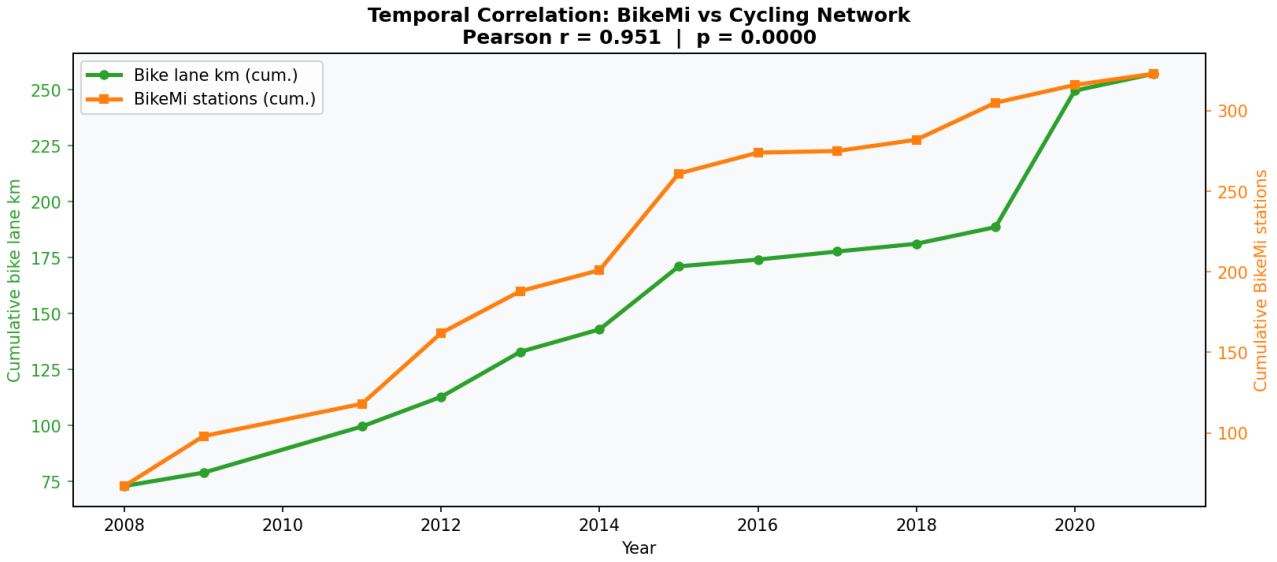


Figure 3: Spatio-temporal correlation between cumulative BikeMi stations and cumulative cycling lane km (2008–2021). Pearson $r = 0.951$, $p < 0.0001$, indicating that the two systems expanded in a highly coordinated manner over time.

The Pearson correlation between the two cumulative time series is $r = 0.951$ ($p < 0.0001$), confirming that BikeMi and cycling lane construction proceeded in tandem at the city level. However, temporal coordination at the aggregate level does not reveal whether the expansion was spatially equitable across income groups — a question addressed in the following sections.

2.3 Spatial Distribution by NIL

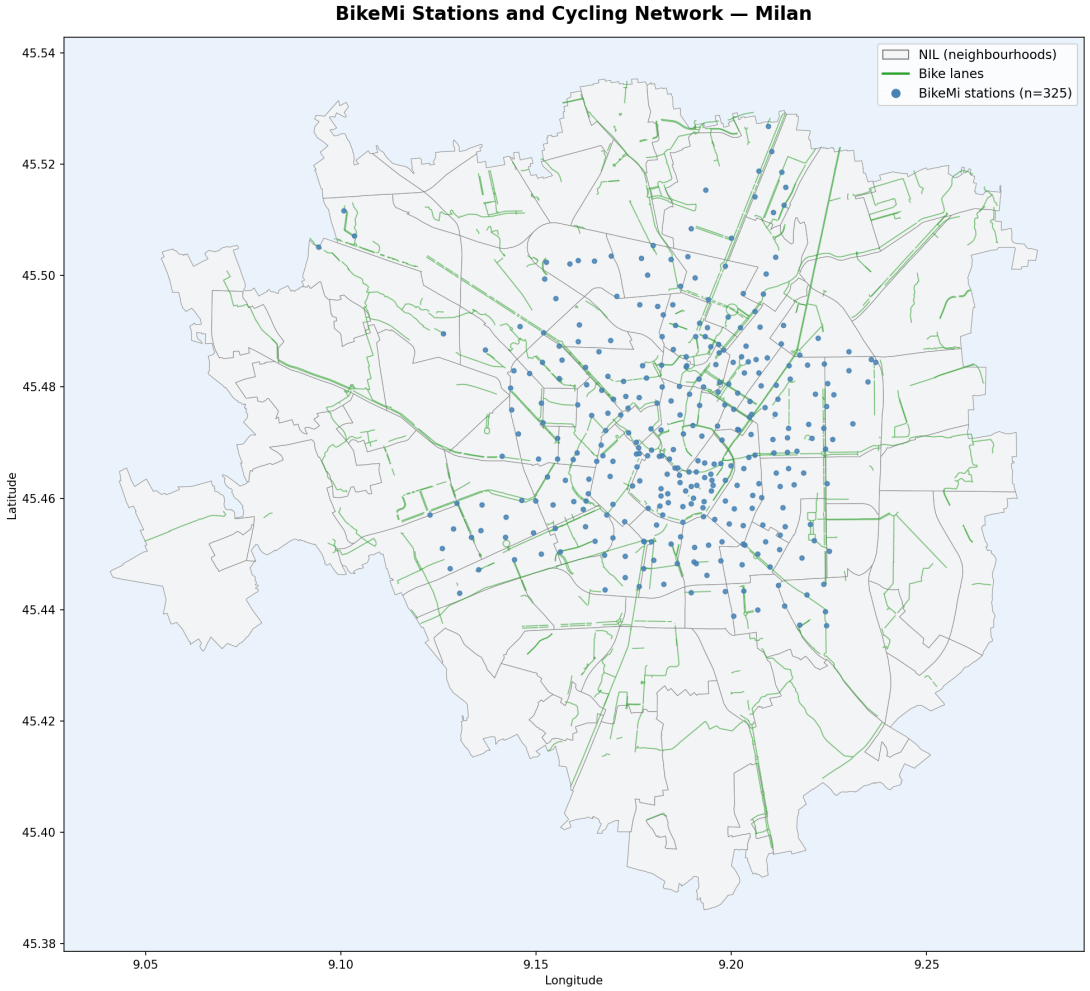


Figure 4: Base map of Milan showing BikeMi stations (blue dots), cycling lanes (green lines), and NIL boundaries. A clear centre-periphery gradient is visible: stations and lanes concentrate heavily in central and inner-ring neighbourhoods.

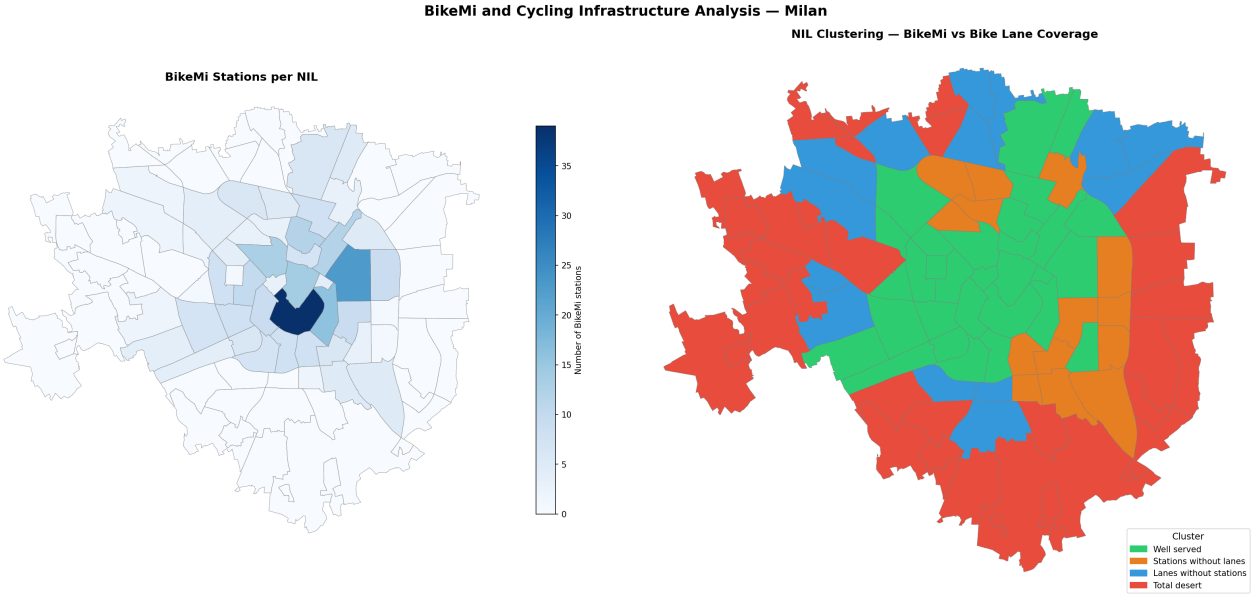


Figure 5: Left: choropleth of BikeMi stations per NIL (darkest: DUOMO with 39 stations, BUENOS AIRES-VENEZIA with 23). Right: 4-type NIL clustering based on the median of station density (0.686 st/km^2) and lane density (1.798 km/km^2). The symmetry between "Well served" and "Total desert" (30 NILs each) confirms a strongly polarised distribution.

The 4-type clustering classifies NILs as follows:

- **Well served**: 30 NILs — above median on both indicators
- **Total desert**: 30 NILs — below median on both indicators
- **Stations, no lanes**: 14 NILs — above-median station density, below-median lane density
- **Lanes, no stations**: 14 NILs — below-median station density, above-median lane density

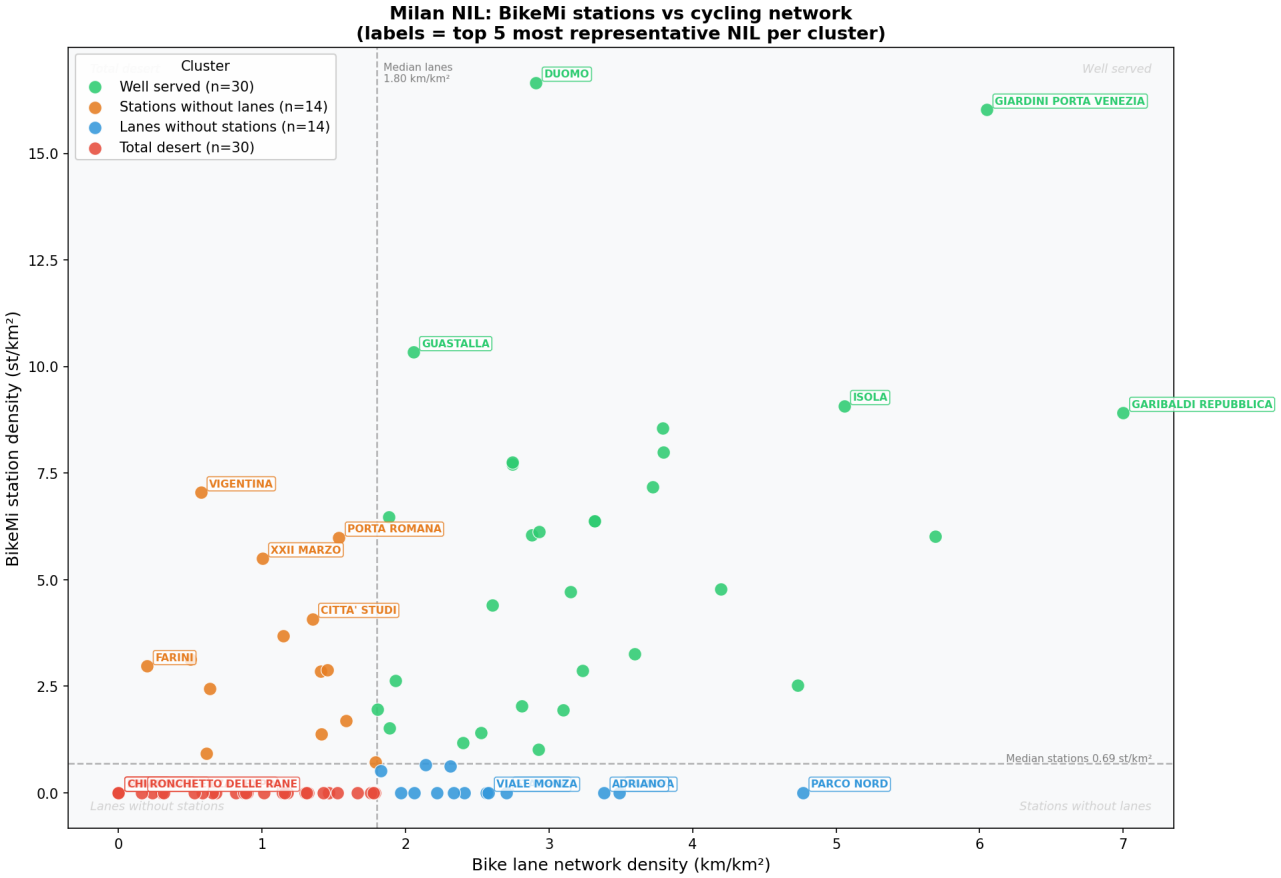


Figure 6: Scatter plot of BikeMi station density vs. cycling lane density per NIL, coloured by cluster. DUOMO (16.7 st/km^2) and GARIBALDI REPUBBLICA are extreme outliers in the "Well served" cluster. CHIRONCHETTO DELLE RANE and VIALE MONZA anchor the "Total desert" cluster in the bottom-left quadrant.

2.4 Pedestrian Accessibility: Isochrone Analysis

For each of the 325 BikeMi stations, the area reachable within 5 minutes on foot at 5 km/h was computed using the actual pedestrian street network of Milan, downloaded from OpenStreetMap via OSMnx and stored locally to avoid repeated downloads.

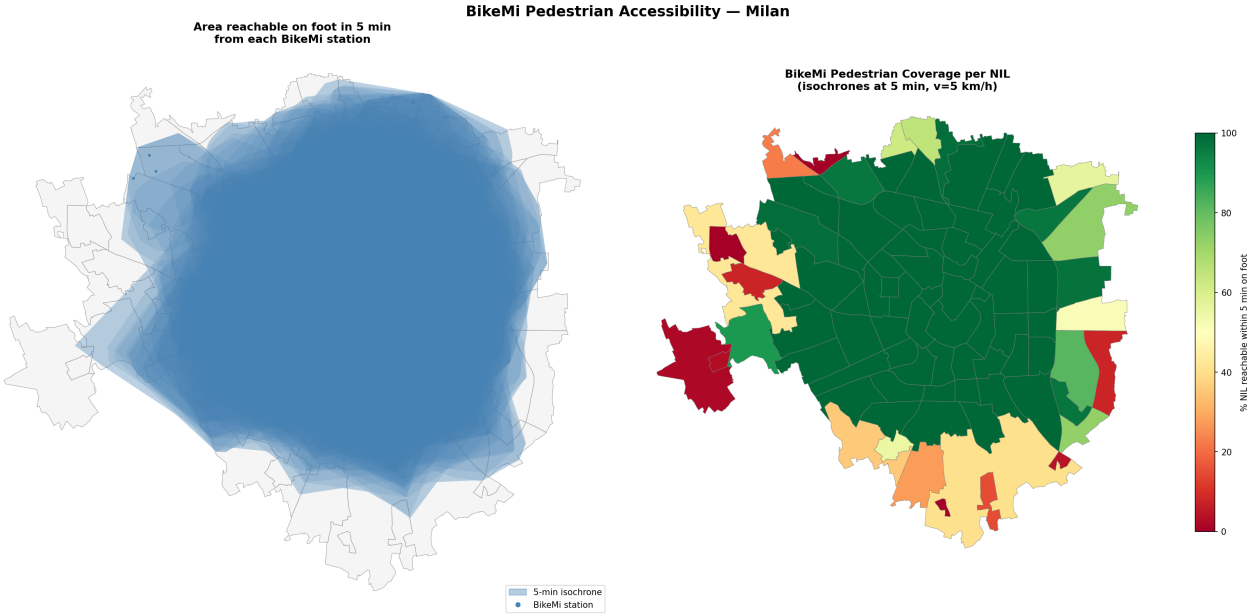


Figure 7: Left: union of 5-minute pedestrian isochrones from all 325 BikeMi stations. Right: choropleth of % NIL area covered within 5 minutes on foot (isochrones at 5 min, $v = 5$ km/h). Average coverage: 82.6%; 8 NILs have coverage below 10%.

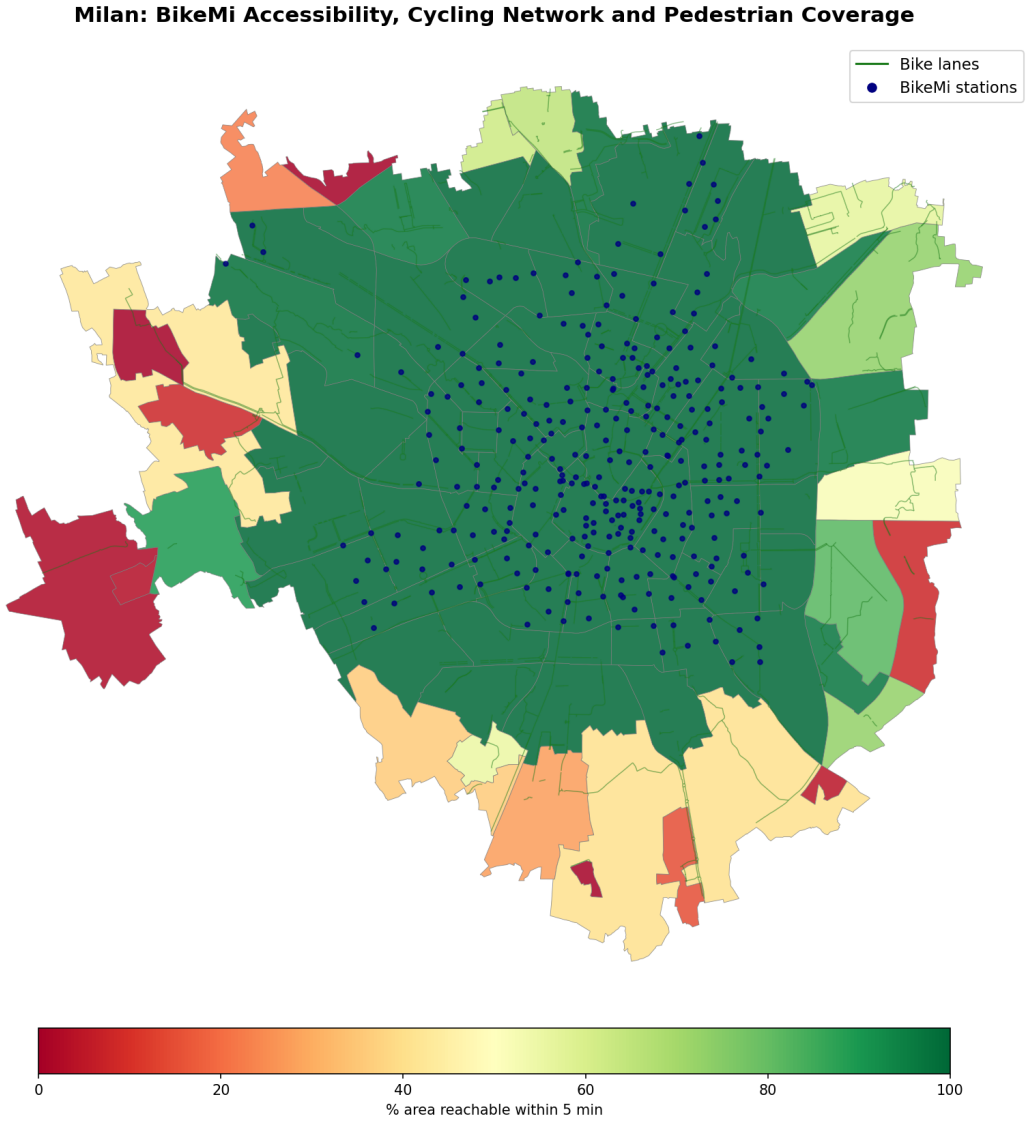


Figure 8: Combined accessibility map: pedestrian coverage choropleth overlaid with cycling lanes (green) and BikeMi stations (navy dots). NILs with coverage below 10%: FIGINO (0%), CHIARAVALLE (0%), RONCHETTO DELLE RANE (0%), SACCO (0.3%), PARCO AGRICOLO SUD (1.9%), MUGGIANO (2.9%), QUINTO ROMANO (6.8%), PARCO MONLUÈ-PONTE LAMBRO (7.5%).

The average 5-minute pedestrian coverage across all NILs is 82.6%, but this average conceals major disparities: 67 NILs achieve coverage above 75%, while 8 NILs have coverage below 10%. These 8 NILs are predominantly peripheral, low-density areas whose residents face near-total absence of BikeMi access.

2.5 Cycling Lane Quality

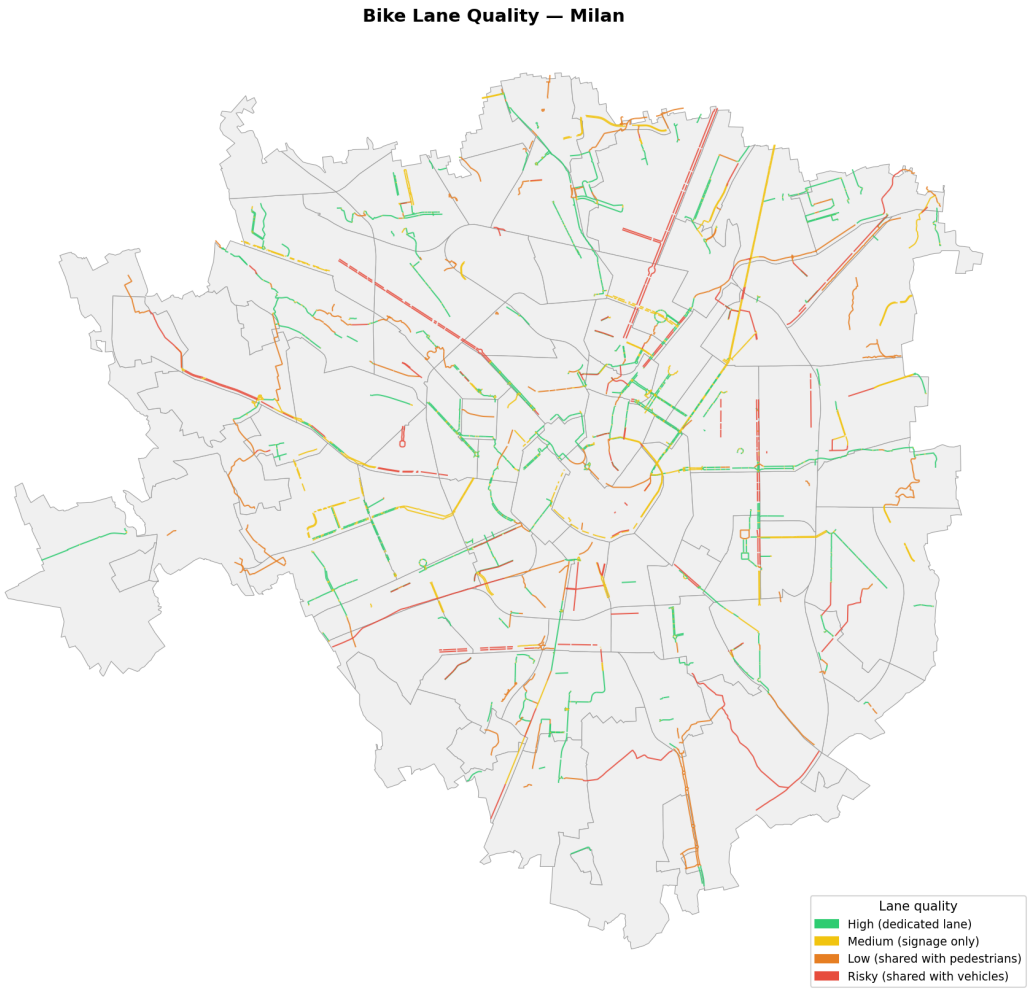


Figure 9: Spatial distribution of cycling lane quality across Milan. Green: dedicated lanes (*ciclabile sede propria*); yellow: signage-only (*ciclabile segnaletica*); orange: shared with pedestrians (*promiscuo pedoni*); red: shared with vehicles (*promiscuo veicoli*).

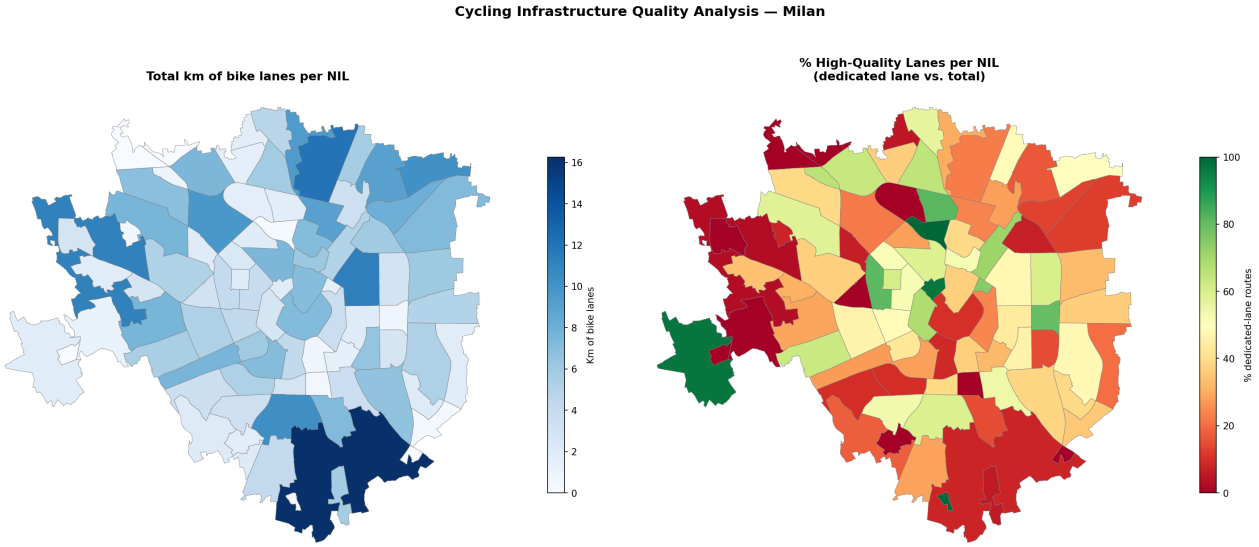


Figure 10: Left: total km of cycling lanes per NIL. Right: % of high-quality dedicated lanes per NIL. The spatial pattern is counter-intuitive: central NILs have the most total infrastructure but the lowest percentage of dedicated lanes, reflecting the historical development of Milan's cycling network.

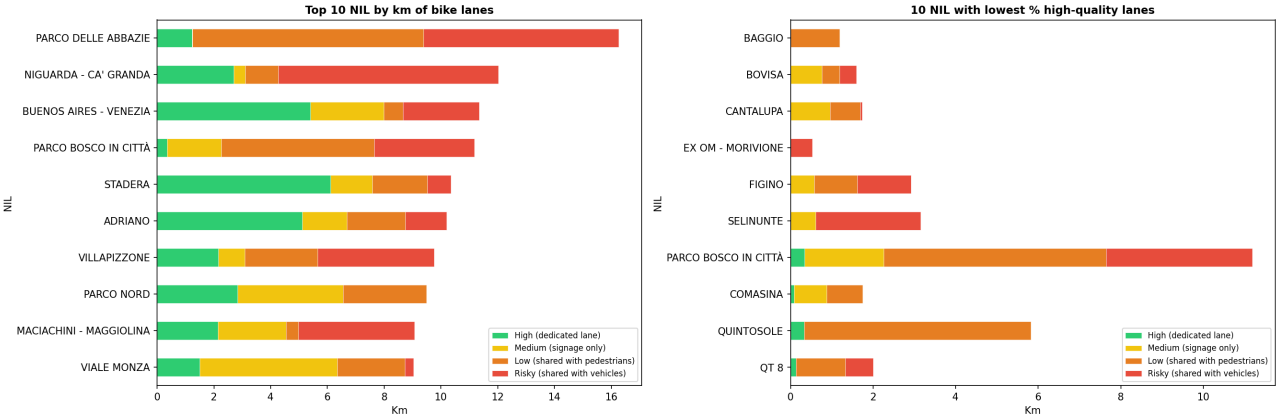


Figure 11: Left: top 10 NILs by total cycling lane km (PARCO DELLE ABBAZIE leads with 16 km). Right: 10 NILs with the lowest % of high-quality lanes, all predominantly peripheral lower-income areas.

Overall, 36.6% of the network consists of high-quality dedicated lanes (*ciclabile sede propria*), 23.8% signage-only, 18.6% shared with pedestrians, and 21.1% shared with vehicles. Only 26 NILs have more than 50% of their lanes classified as high-quality.

2.6 Socioeconomic Equity Analysis

2.6.1 Income Distribution

Figure 12: Left: average IRPEF income per NIL (Comune di Milano, tax year 2022). The centre-periphery gradient is pronounced: BRERA (€94,369), DE ANGELI-MONTE ROSA (€83,768), MAGENTA-S.VITTORE (€74,232). Lowest: QUINTO ROMANO (€18,509), QUARTO OGGIARO (€23,602). Right: 4-cluster mobility map for comparison.

2.6.2 Correlation Results

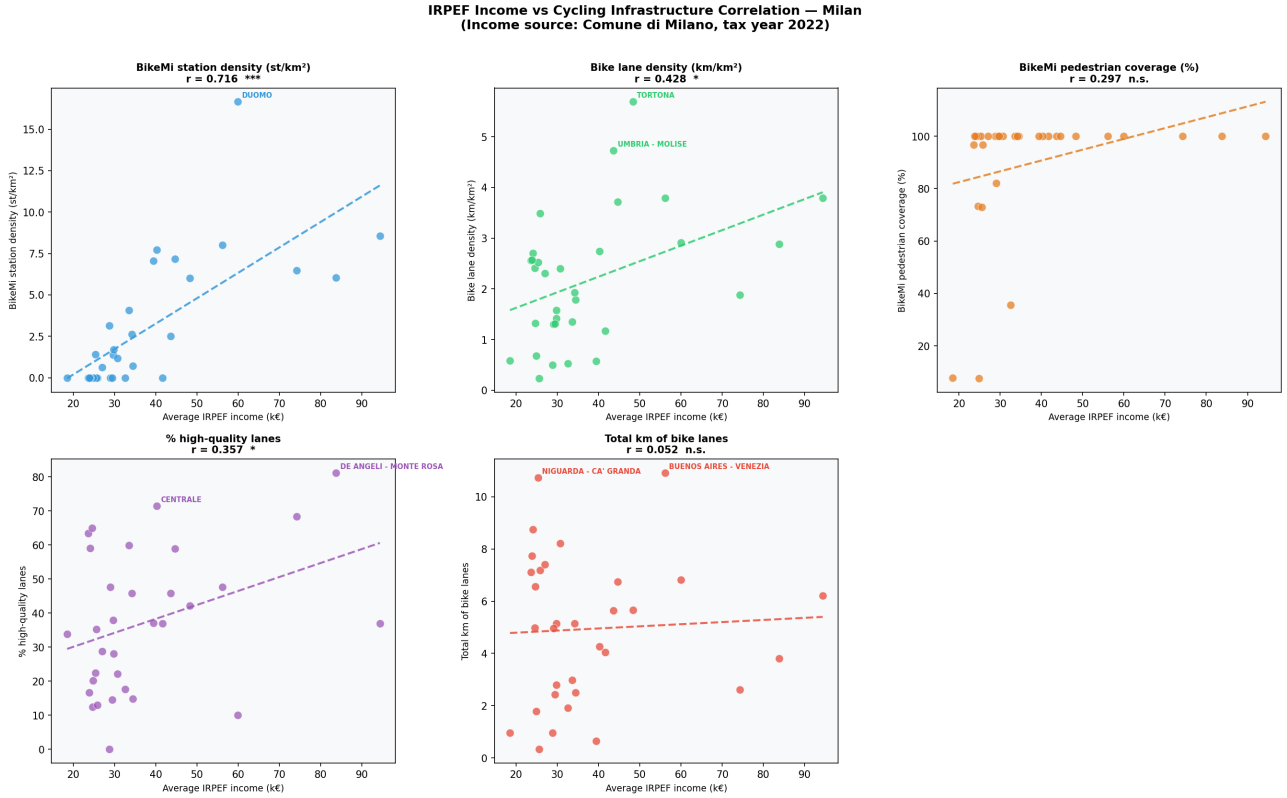


Figure 13: Scatter plots of average IRPEF income (k€) vs. each accessibility indicator, with linear regression fit (dashed line). The strongest correlation is with BikeMi station density ($r = +0.716$, $p < 0.001$). Total cycling lane km shows no significant correlation ($r = +0.052$, n.s.).

Table 1 summarises the Pearson correlations between NIL income and each accessibility indicator ($n = 32$ NILs with complete data).

Table 1: Pearson correlations between average IRPEF income and cycling accessibility indicators ($n = 32$ NILs with complete data).

Indicator	**Pearson r**	**p-value**	**Significance**
BikeMi station density (st/km^2)	+0.716	<0.001	***
Cycling lane density (km/km^2)	+0.428	0.015	*
Pedestrian coverage (%)	+0.302	0.093	n.s.
% high-quality lanes	+0.357	0.045	*
Total km cycling lanes	+0.052	0.778	n.s.

*** $p < 0.001$; * $p < 0.05$; n.s. not significant

The correlation between income and BikeMi station density ($r = +0.716$, $p < 0.001$) is the most striking result: wealthier NILs systematically have more BikeMi stations per km^2. Notably, the total km of cycling lanes does not correlate with income ($r = +0.052$, n.s.), meaning the city does invest in cycling infrastructure across all income levels — but it builds better quality infrastructure (dedicated lanes) and places more BikeMi stations in higher-income areas.

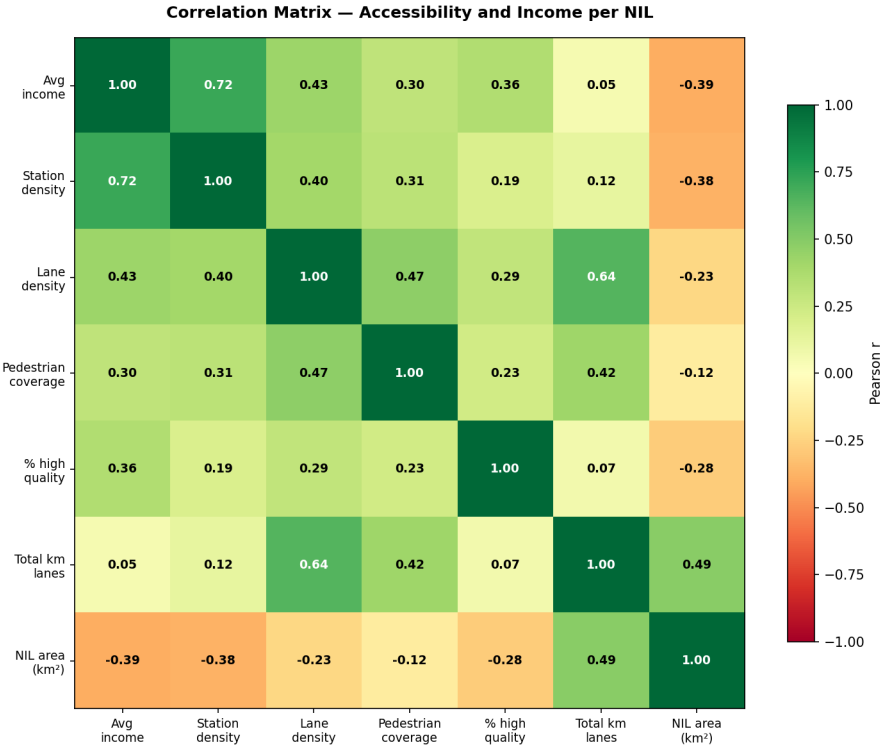


Figure 14: Full Pearson correlation matrix across all indicators and NIL area. Station density and lane density are moderately correlated ($r = 0.40$). NIL area correlates negatively with both density indicators, as expected for larger peripheral neighbourhoods.

2.6.3 Income Distribution by Cluster

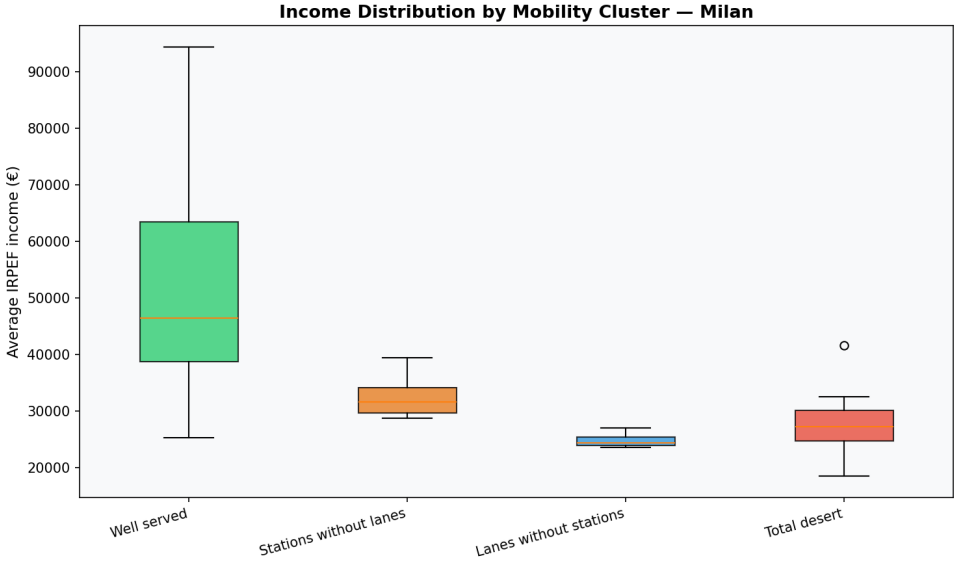


Figure 15: Distribution of average IRPEF income by mobility cluster. "Well served" NILs: median ≈€47,000 (range €24,000–€94,000). "Lanes, no stations": median ≈€24,000, the lowest and most homogeneous cluster. The income gap between "Well served" and "Total desert" averages ≈€20,000 per taxpayer per year.

These results confirm that the allocation of BikeMi stations follows a clear socioeconomic gradient, consistent with findings from the international literature [6, 7] and with a parallel study on Milan's broader transport accessibility [13], which found $r = +0.718$ between income and a composite Milan Accessibility Index. This convergence across independent methodologies strengthens the validity of the finding.

2.7 Classification Model: Decision Tree for Priority NILs

2.7.1 Methodology

A Decision Tree classifier was trained to identify NILs that should be prioritised for new BikeMi station investments. The target variable was defined as binary:

- **Priority (1)**: NILs classified as "Total desert" or "Lanes, no stations" — areas with insufficient bike-sharing access given their existing cycling infrastructure
- **Not priority (0)**: NILs classified as "Well served" or "Stations, no lanes"

This labelling reflects the policy question: given existing cycling infrastructure, where should new BikeMi stations be added to maximise equity? Five features were used: average IRPEF income (€), NIL area (km^2), cycling lane density (km/km^2), percentage of high-quality lanes, and pedestrian isochrone coverage (%). The model was evaluated using 5-fold stratified cross-validation with grid search over depths 2, 3, and 4.

2.7.2 Results

The optimal depth was 2 (CV accuracy = 0.790 ± 0.142), substantially above the random baseline of 50% for a balanced binary classification. The model learned two interpretable decision rules:

```
IF avg_income <= EUR 29,557:
    IF % high-quality lanes > 6.2%  ->  PRIORITY
    ELSE                             ->  NOT PRIORITY
IF avg_income > EUR 29,557:
    IF NIL area > 3.43 km2          ->  PRIORITY
    ELSE                             ->  NOT PRIORITY
```

The first rule captures low-income NILs: if they have some cycling infrastructure (quality > 6.2%), adding BikeMi stations is warranted. The second rule captures large higher-income peripheral NILs where station density is low due to size. Both rules are directly interpretable by policymakers without requiring specialist modelling knowledge.

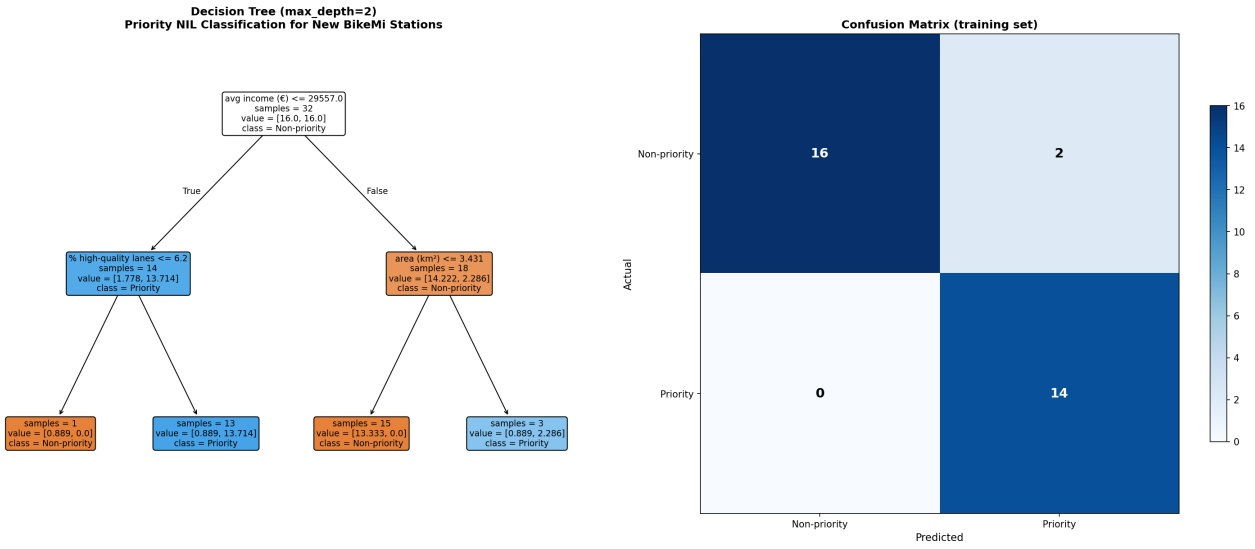


Figure 16: Left: Decision Tree visualisation (max_depth = 2). Right: confusion matrix on the training set. 16 correct "not priority" predictions, 14 correct "priority" predictions, 2 false negatives (missed priority NILs), 0 false positives.

2.7.3 Priority NIL Map

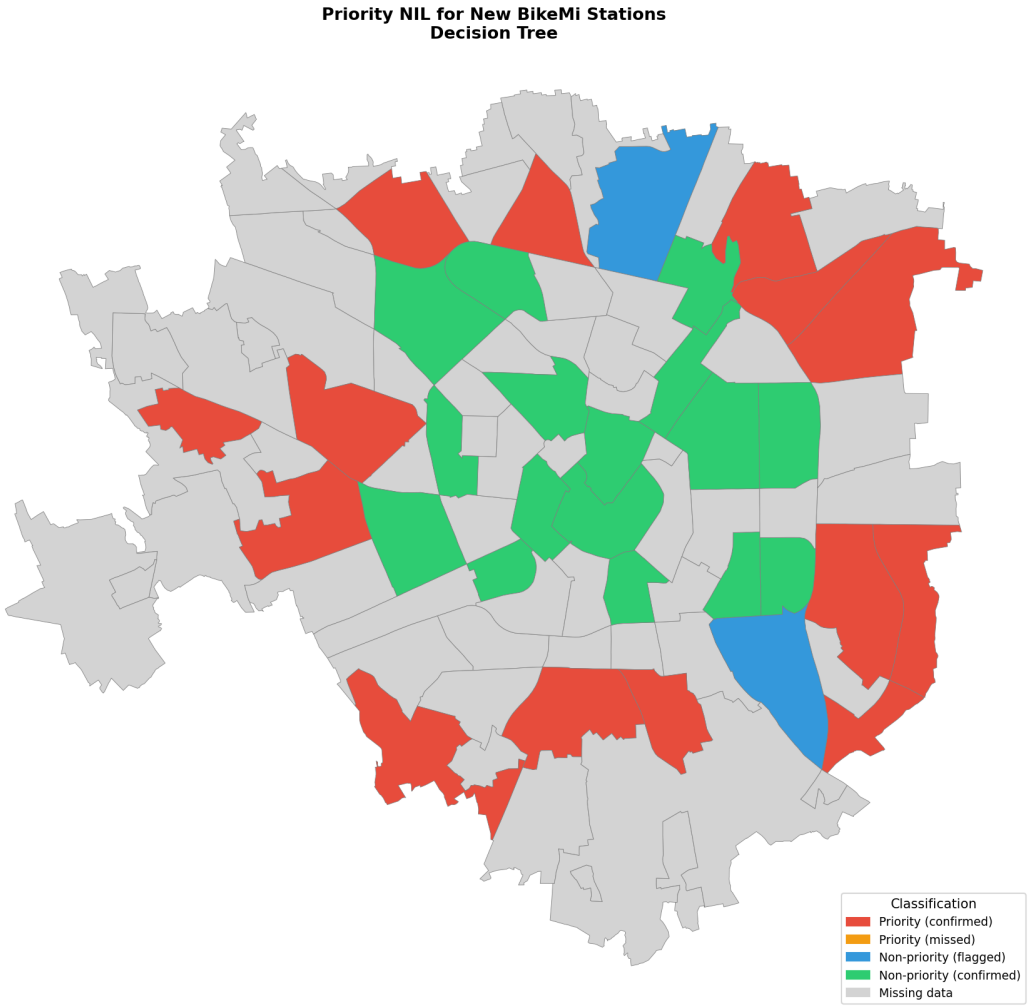


Figure 17: Priority NILs for new BikeMi stations identified by the Decision Tree classifier. Red: priority confirmed; green: not priority confirmed; blue: non-priority flagged by model; grey: missing data. The 16 priority NILs are predominantly located in the northern, western, and southern periphery of Milan.

The 16 priority NILs identified are: PARCO LAMBRO-CIMIANO, S. SIRO, QUARTO OGGIARO, AFFORI, PADOVA, TRIULZO SUPERIORE, PARCO MONLUÈ-PONTE LAMBRO, NIGUARDA-CA' GRANDA, STADERA, LODI-CORVETTO, QUINTO ROMANO, MECENATE, RIPAMONTI, VIALE MONZA, FORZE ARMATE, and PARCO DEI NAVIGLI.

2.8 Policy Recommendations

Recommendation 1 — Prioritise BikeMi expansion in the 16 identified NILs. The Decision Tree model identifies 16 NILs where the combination of low income, existing cycling infrastructure, and absence of bike-sharing coverage creates a strong equity case for investment. QUARTO OGGIARO (€23,602), QUINTO ROMANO (€18,509), and VIALE MONZA (€23,820) have the lowest incomes and should receive highest urgency, with a minimum of 3 stations per NIL placed on existing cycling lanes.

Recommendation 2 — Upgrade cycling infrastructure quality in low-income peripheral NILs. The 10 NILs with the lowest percentage of dedicated cycling lanes (BAGGIO, BOVISA, CANTALUPA, EX OM-MORIVIONE, FIGINO, SELINUNTE, PARCO BOSCO IN CITÀ, COMASINA, QUINTOSOLE, QT8) require qualitative upgrading — replacing shared-vehicle and shared-pedestrian lanes with physically separated infrastructure. Priority should be given to *promiscuo veicoli* segments (21.1% of the network) which pose direct safety risks.

Recommendation 3 — Introduce equity-adjusted BikeMi pricing. Eight NILs have near-zero pedestrian coverage from BikeMi stations, meaning that adding stations is not sufficient — residents may also face affordability barriers. Milan could follow the example of Chicago's "Divvy for Everyone" programme and New York's "reduced-fare membership" by introducing income-adjusted annual passes for residents of identified low-income NILs [7].

Recommendation 4 — Establish a cycling equity monitoring framework. The five indicators developed in this study (station density, lane density, pedestrian coverage, lane quality, income) should be institutionalised as an annual monitoring dashboard, enabling the city to track whether successive BikeMi expansions and cycling infrastructure investments are reducing or widening the equity gap. The Decision Tree model provides a reproducible classification methodology that can be updated annually.

2.9 Discussion and Limitations

The central finding — a Pearson correlation of $r = +0.716$ ($p < 0.001$) between NIL income and BikeMi station density — is consistent with the international literature on bike-sharing equity [6, 7] and with the parallel study on Milan's transport accessibility [13], which found $r = +0.718$ between income and a composite accessibility index. This convergence across independent methodologies strengthens the validity of the finding.

Several limitations should be acknowledged. First, income data is available at CAP level (38 postal codes) rather than NIL level (88 neighbourhoods), requiring spatial interpolation via centroid-based assignment, which introduces measurement error. Second, the analysis is cross-sectional (tax year 2022), preventing causal inference. Third, the Decision Tree was trained on $n = 32$ NILs, a small sample that limits statistical power and may produce unstable decision boundaries.

Despite these limitations, the consistency of findings across multiple indicators and analytical approaches provides a robust evidential basis for the policy recommendations.

2.10 Ethical and Social Implications of the Solutions

The policy recommendations involve explicit prioritisation of lower-income neighbourhoods for public infrastructure investment. Positive prioritisation is consistent with the vertical equity principle [11] and Milan's PUMS commitments, but requires transparent justification. The use of objective, data-driven indicators (income, cycling infrastructure quality, pedestrian coverage) as the basis for prioritisation provides this justification.

The use of income data in the classification model raises privacy concerns: while the IRPEF data is aggregated at postcode level and published by the Comune di Milano, its use in automated infrastructure decisions should be subject to transparency and accountability mechanisms. Residents of priority NILs should be informed of and involved in the planning process for new stations.

Finally, there is a risk that expanding BikeMi into currently underserved neighbourhoods could accelerate gentrification, as documented in some US cities [14]. Policymakers should monitor housing prices in priority NILs following any BikeMi expansion and consider complementary anti-displacement policies.

GenAI Usage Statement

This essay was produced with the support of Claude (Anthropic) as a coding assistant for the implementation of data wrangling, spatial analysis, and visualisation code in Part 2 (Google Colab / Python). All analytical decisions, interpretation of results, and textual content were produced by the authors. The code was fully reviewed, understood, and is presentable and explainable by both authors. GenAI was not used to generate the textual content of the essay.

CRediT Author Statement

Davide Fabio Loreti: Conceptualisation, Data Curation, Software (Cells 1–4), Writing — Original Draft (Part 2, Sections 2.1–2.5).

Carlo Pegoraro: Conceptualisation, Formal Analysis, Software (Cells 5–6), Writing — Original Draft (Part 1 and Sections 2.6–2.10), Visualisation.

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